

Meeting Notes

Date

Jan 20, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan
- @Hussein Khodor

Goals

- Meet with our facility advisor
- Discussion of Analog Devices BMS Products

Discussion topics

Time	Topic	Presenter	Notes
15:40	Analog Devices BMS Products	All participated	• Found a BMS “dev kit” called AD-CELLPACKBM-SL • AD-CELLPACKBM-SL contains Devices that can simulate battery pack cells.
14:25	Unable to meet with Advisor	N/A	• Move Advisor meeting to Thursday on 15:30

Date

Jan 22, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan

- @Hussein Khodor
- Facility Advisor: Andrew GreenburGoals
- Battery Talk with Facility Advisor

Discussion topics

Time	Topic	Presenter	Notes
15:30	Battery Talk	Andrew Greenburg	• Set Pack Voltage • I^2R determines wire size • High Voltage safety • FSAE Rule EV 3.3.2 600 V DC->Charger overhead→Battery Pack 504 V→Nominal Voltage 446 V • Guaranteed Maximum Safe Voltage for Humans: 48 V • Maximums for Modules (Grouping of Cells FSAE EV 5.1.1): Weight-12 kg, Static DC Voltage 120 V, and Energy 6 MJ • BMS uses Isolated Protocol for measuring cells to avoid 600V

Action items

- ☒ Create a “Rulebook Requirements” Document @David Fransis
- ☐ Choose Lithium-Nickel-Manganese-Cobalt or Lithium-Iron-Phosphate
- ☐ Research Isolated Communication Protocols and CAN Bus

Decisions

 Lithium Battery Chemistry

Related info

Date

Jan 30, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan
- @Hussein Khodor
- Industry Sponsor: Jake and Antonio

Goals

- Discuss our research on battery chemistry
- Decide which chemistry to use

Discussion topics

Time	Topic	Presenter	Notes
19:05	Catch up	Jake and Antonio	• Charging during race: Doesn't happen • Old battery pack was 200 V, • Peak Current • Create small prototypes with documentation for full design and modularity •

Action items

☐

Decisions

↶

Related info

Date

Feb 6, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan
- @Hussein Khodor
- Faulty Advisor: Andrew Greenberg

Goals

- Discuss options for system architecture

Discussion topics

Time	Topic	Presenter	Notes
15:45	Using Open Source BMS	@Hussein Khodor and @Fangdong Yuan	• Distributed Architecture: Mini BMSs with module(s) • No interloper: Don't bother with compiling with a separate chip. Using the monitoring and fault data is the VCU's job. • Look into Battery Cutoff • Analog Protection: No Microcontroller required

Action items

- ☐ Order TI Battery Monitoring IC found in Libre Solar BMS
- ☐ Find evaluation boards to connect our ICs for testing
- ☒ Update L1 Block Diagram

Decisions

👉 Distributed Architecture

👉 Leave data processing for VCU

Related info

Date

Feb 10, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan
- @Hussein Khodor
- Allen P.
- Jack
- Estefani

Goals

- Touch base with VCU/ECU team
- Remind ourselves what to deliver by the end of the week

Notes

- Distributed Architecture must interface with CAN bus with Singler module
- The Battery Control Unit (BCU) will conform with VCU standards for satellite modules

Action items

- ☐ READ THE RULES!!!!

Decisions

👉 Distributed Architecture through BCU

👉 Use ZEPHYR 4.3 for BCU

👉 Use 511-NUCLEO-C092RC for BCU

Related info

Date

Feb 13, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan
- @Hussein Khodor
- Allen P.
- Jack
- Estefani
- Faulty Advisor: Andrew Greenberg

Goals

- Touch base with VCU/ECU team
- Fill WPR and IC BOM

Notes

- CAN Database
 - Messages: Class of signals
 - Signals: The specific format
- Don't **assume** unidirectional communication
- Found new TI IC: BQ79616
 - Can daisy chain through
 - Calculates Sensor Data

Action items

- ✓ Determine Message Definitions to be send the VCU (write down into google doc)

Decisions



Date

Feb 17, 2026

Participants

- @Jose Ramirez
- @David Fransis

- @Fangdong Yuan
- @Hussein Khodor

Goals

- Touch base with VCU/ECU team
- Fill WPR and IC BOM

Notes

- BQ79656 was decided for the monitoring chip since it allow for earlier development of firmware to allow for parallel engineering.
- BMS and VCU will be powered with 12 V battery

Action items

- ✓ Send Andrew the component list through email

Decisions

- ✓ BQ79656 as monitoring chip for now

Date

Feb 27, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan
- @Hussein Khodor
- Allen P.
- Jack
- Estefani
- Faulty Advisor: Andrew Greenberg

Goals

- Touch base with VCU/ECU team
- Fill WPR

Notes

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Action items

- ☒ Determine Message Definitions to be send the VCU (write down into google doc)

Decisions



Date

Mar 4, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan
- @Hussein Khodor

Goals

- Plan TED Talk for Friday
- Documentation for System Hardware

Notes

- We'll present with google slides
 - Each bullet point on assignment description will be a slide or half a slide
 - @Hussein Khodor will start off, @David Fransis will cover what we have delivered, @Fangdong Yuan will talk about System Architecture overview, @Jose Ramirez Will talk about what' good and what's not

Action items

- ☒ Create google slide for the presentation
- ☐ Use Hierarchical Sheets to KiCAD Schematic

Date

Mar 6, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan
- @Hussein Khodor
- Jack
- Louis
- Faulty Advisor: Andrew Greenberg

Goals

- Touch base with VCU/ECU team
 - Talk about how the BMS should connect to daisy chain CAN-FD Bus
 - CAN Protocol
- Talk about Component order with Andrew
- Ask Andrew about the design
- Fill WPR

Notes

- When Battery is unstable, BMS sends Fault Message to VCU and activates Shutdown circuit
- VCU will provide:
 - Arbitration Identifier
 - Transmission specification with App file: contains constants Frequency of CAN Messaging
 - CAN Development Document
 - Daisy Chain Harness CAN Bus (CANH, CANL, GND)
- Use email to contact Andrew with Important messages
- Parts ordered during meeting
- **Include Isolation Module into prototype by end of term**

Action items

- ☐ Read more of the Zephyr Manuals
- ☐ @Fangdong Yuan Label all the lines in the block diagram

Decisions

✔ Include Isolation module into prototype

✔ Provide voltage to the LV level

Date

Mar 10, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan
- @Hussein Khodor
- Industry Sponsor: VMS - Jake
- Faulty Advisor: Andrew Greenberg

Goals

- Inform Industry sponsor about components
- Ask how exactly we would send them order forms
- Add small tweaks to the Block Diagram
- Update on the KiCAD Schematic

Notes

- Communication isolation module already on Monitoring chip so communication can bypass the Digital Isolation Unit
- David has looking into ISO and AIR ICs
- Firmware development can be done through either VS IDE or text & terminal
- Faculty Advisor and Industry Sponsor representative didn't show up
- Monitoring IC will have a separate PCB that will connect to the main BMS
 - Allows for easier scalability
- We're designing system architecture, **not the evaluation boards or the ICs**
- UART Communication can through DMA in Zephyr

Action items

- ☐ @David Fransis List ISO and AIR Components
 - ☐ Below all the listings include lists of criteria for each component type

Decisions



Date

Mar 20, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan
- @Hussein Khodor
- Faulty Advisor: Andrew Greenberg

Goals

- Format plan for firmware development with VCU team
- Check up on System Architecture Development
- Inquire about IC components

Notes

- Satellite development, not much firmware
- Firmware goal: Deliver a CAN message of a voltage reading
- Order of ICs confused grant and was lost; Must **reorder**
- Deciding having Current Sensor integrated to chip or as separate board for more modular battery pack
- Kevin Connections - op-amp across high voltage resistor

Action items

- ☐ @Fangdong Yuan Begin Development on Cell Monitoring and Current Sensor Designs
- ☐ Write up of a current Shunt
 - ☐ What it is
 - ☐ Tradeoffs
 - ☐ Final decision for current sender

Decisions

🌱 Current Sensor Separated from monitoring IC

Date

Apr 2, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan
- @Hussein Khodor
- Industry Sponsor: VMS - Jake
- Faulty Advisor: Andrew Greenberg

Goals

- Firmware Development
- Update on the Architecture Plan

Notes

- Battery Monitoring Module Schematic mostly complete
- Components for Firmware development have arrived

Action items

- ☐ @David Fransis @Fangdong Yuan Finish Schematics by Friday
- ☐ @Hussein Khodor Complete UART driver to Communication Bridge by Monday
- ☐ @Jose Ramirez Set up CAN shell on Nucleo Boards by Monday

Decisions

📌 Monitor only pack temperature (Optional Module Temperature Readings)

Participants

- @Jose Ramirez
- @Fangdong Yuan
- @Hussein Khodor

Goals

- Write up TED Talk
- Plan for Friday

Notes

- Clarification on BMS CAN FD Message Sequencing
 - Men

Action items



Decisions



Date

Apr 2, 2026

Participants

- @Jose Ramirez
- @David Fransis
- @Fangdong Yuan
- @Hussein Khodor
- Industry Sponsor: VMS - Jake
- Faulty Advisor: Andrew Greenberg

Goals

- Firmware Development
- Update on the Architecture Plan

Notes

- Battery Monitoring Module Schematic mostly complete
- Components for Firmware development have arrived

Action items

- ☐ @David Fransis @Fangdong Yuan Finish Schematics by Friday
- ☐ @Hussein Khodor Complete UART driver to Communication Bridge by Monday
- ☐ @Jose Ramirez Set up CAN shell on Nucleo Boards by Monday

Decisions



Monitor only pack temperature (Optional Module Temperature Readings)

Date

Apr 3, 2026

Participants

- @Jose Ramirez
- @Fangdong Yuan
- @Hussein Khodor
- @David Fransis
- Faculty Advisor

Goals

- Schematic Design Review
- Project Plan revision

Notes

- Design Review
 - Make more readable
 - Make TSRef into power supply with a voltage supply

Action items



Decisions



Date

Apr 6, 2026

Participants

- @Jose Ramirez
- @Fangdong Yuan
- @Hussein Khodor
- @David Fransis
- Faculty Advisor

Goals

- Schematic Design Review (Full)
- Firmware Development Update
- Project Re-planning and De-scoping

Notes

- Unfortunately, our facility advisor wasn't able to make it.
- Cutting down the shoulds and mays of the project expect for the modular architecture
- Full L2 diagram of BMU
- IMD handles isolation faults

Action items

- ☐ @Fangdong Yuan Add revisions to the BMU schematic by Friday

Decisions

- 👉 Will on only deliver on the Musts of the Capstone project

Date

Apr 13, 2026

Participants

- @Jose Ramirez
- @Fangdong Yuan
- @Hussein Khodor
- @David Fransis
- Faculty Advisor
- Industry Sponsor

Goals

- Battery Requirements from VMS
- Software Block Diagram
- Hardware Development Check up

Notes

- New Current Monitoring IC

- Can create simple over-current protection
- Uses TI I2C FAST_MODE
- Battery Requirements
 - Motor Specs

🔖 [Figure 6.1 & 6.2 - Motor core and exploded view](#)

6.1.1 Description, type, operating parameters

The Remy HVH 250 HT is the high torque (110mm core) version of their oil cooled (ATF) motor, used in numerous hybrid electric cars. The motor is rated for 87KW peak power at an input voltage of 320 volts. This application runs at a reduced voltage (190V) and power rating (42kW).

Motor Manufacturer and Type:	Remy HVH 250-115 HT
Motor principle	permanent magnet AC
Maximum continuous power:	120 kW
Peak power:	160kW
Input voltage:	320 VAC
Nominal current:	200 A
Peak current:	300 A
Maximum torque:	330 Nm

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Nominal torque:	248 Nm
Cooling method:	Oil-injected

- Old pack: ~300V NMC Li Ion
- Motor Controller Specs

 [Figure 5.1 - Picture of motor controller](#)

[5.1.1 Description, type, operation parameters](#)

The motor is operated by a [RMS](#) PM100 motor controller. The motor has [software](#) onboard that varies the voltage supply to the motor based on input directly from the motor, as well as CAN messages from the vehicle [control unit](#).

Motor controller type:	RMS PM100DX Controller
Maximum continuous power:	105 kW
Maximum peak power:	122 kW
Maximum Input voltage:	360 VDC
Output voltage:	187 VAC
Maximum continuous output current:	300 A
Maximum peak current:	350 A for 5s
Control method:	CAN, Hardware I/O
Cooling method:	H2O cooled via radiator
Auxiliary supply voltage:	12 VDC

 [Table 5.1 General motor controller data](#)

- Sponsor Requests
 - Presentations
 - Shutdown signal
 - Digital logic from STM32

Action items



Decisions



Date

Apr 24, 2026

Participants

- [@Jose Ramirez](#)
- [@Fangdong Yuan](#)
- [@Hussein Khodor](#)
- [@David Fransis](#)
- Faculty Advisor

Goals

- Check In with VCU team
 - CAN Database revisions
- Software Block Diagram
- Hardware Development Check up

Notes

- Design Review Notes
 - Fix D2 and Q2 in wrong
 -
- No one cell should have more than
- BMS CAN Messages are determined by us
- Demo car: 12V, max current 30 A
- Prototype

Action items

- ☐ @Fangdong Yuan Implement Design Review Changes
 - ☐ Fix D2 and Q2 to right direction
 - ☐ Finish BMU COM Bridge schematic

Decisions

57

Date

Apr 27, 2026

Participants

- @Jose Ramirez
- @Fangdong Yuan
- @Hussein Khodor
- @David Fransis
- Faculty Advisor

Goals

- Software Re-planning
- Hardware Development Check up

- Friday Demonstration Plan

Notes

- Software
 - @Jose Ramirez Has been struggling to keep up with classes and needs help with CAN code
 - @Hussein Khodor Also has full course load
 - CAN functionality will have to be put off to next week
- Hardware
 - Schematics Finished
 - PCB layouts started
 - BMU v1.0 Draft Complete
 - SHUNT and monitoring modules difficulties
 - KiCAD schematic import not working
- Demonstration
 - Voltage and Temperature monitoring ready
 - Use a 24V, 5W halogen bulb for the Battery Load
- Documentation
 - Test Plan
 - For June Implementation

Action items

- ☐ @Jose Ramirez

Decisions

🔄 CAN-FD Delayed

Date

May 4, 2026

Participants

- @Jose Ramirez
- @Fangdong Yuan
- @Hussein Khodor
- @David Fransis

- Faculty Advisor

Goals

- Software Re-planning
- Hardware Development Check up
- Discuss what should be included in user's and developer's manuals

Notes

- Software
 - Fault Configuration and Reading Complete
 - Cell Balancing and SHUNT module code
 - CAN test message **will** be done by Friday
 - CAN packaging and Transmitting for main project will be done
- Hardware
 - "Voltage sense Overcurrent Protection device(s) blown or tripped" is below the threshold the HV fuse is blown
- Demonstration
 - Separate Batteries for different voltage system
- Documentation
 - User's Manual
 - How to reset the BMS after a fault was detected
 - Look for which fault
 - Developer's Manual
 - Assembling and scaling the battery pack
 - Minor adjustments to software constants for pack scaling

Action items



Decisions

